Contents- Unit-I **Distributed Ledger Technology**

Distributed Ledger Technology:

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  - Process flow
- Institutional design elements
  - Operation of the arrangement
  - Access to the arrangement (unrestricted or restricted)
- Potential configurations and trade-offs

Contents- Unit-II **Analytical framework**

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Four main component of framework

Framework I --- Rules and Structure while

Analytical framework is the "evaluation and improvement method."

Aspect	Framework	Analytical Framework
Definition	Structure, rules, and mechanisms that define how a system operates.	A method to analyze and evaluate how well the system performs.
Purpose	Establishes how a system functions securely and efficiently.	Assesses performance, security, and improvements.
Real-Life Example (Traffic System)	Traffic laws, signals, and road rules that regulate driving.	Evaluating traffic congestion, accident rates, and road efficiency to improve city planning.
Real-Life Example (DLT)	Bitcoin’s PoW consensus, block structure, and decentralized governance.	Measuring Bitcoin’s transaction speed, security, and energy consumption to assess efficiency.

DLT Framework vs. Analytical Framework

DLT framework defines the rules and structure of a distributed ledger

including protocols, consensus mechanisms, and governance.

For example Bitcoin's framework consists of:

Consensus: Proof of Work (PoW) for transaction validation.

Data Structure: Blocks forming a chain.

Governance: Decentralized decision-making by nodes.

Where as analytical framework evaluates a DLT system's performance, security, and efficiency.
For example, analyzing Ethereum based on:

Scalability: Transaction speed.

Security: Resistance to attacks.

Decentralization: Node distribution.

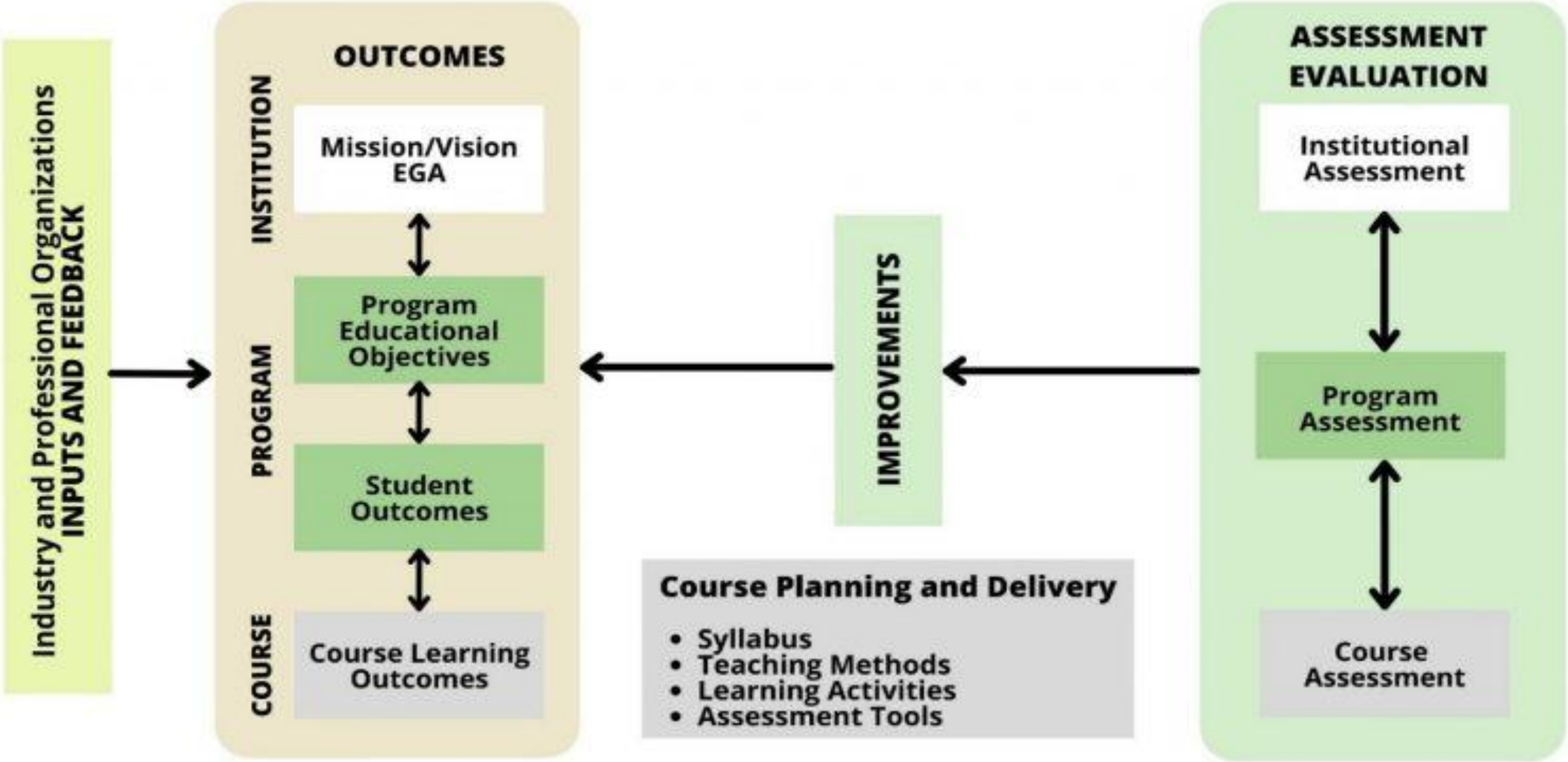
Energy Efficiency: PoS vs. PoW energy consumption

Framework --"how it works," while Analytical framework --"how well it works."

Framework

Purpose of a framework :-

- To assist in development and Providing standard
- Take Care of Low-level functionality
- To serve as a support/guide for building of something that expands the structure into something useful.
- It serves as a foundation, so need not to start entirely from scratch.
- One type of tool used by an organizations to support their process and organizational improvement efforts



Framework for academic program assessment and continuous improvement

Inputs & Feedback(Left Side**):** Industry and professional organizations provide guidance.

Outcomes(Middle Section**):**

Institutional: Mission/Vision

Program: Educational Objectives & Student Outcomes

Course: Course Learning Outcomes

Assessment & Evaluation(Right Side)**:**This section ensures **program effectiveness**:

Course, Program, and Institutional Assessments ensure effectiveness.

Course Planning & Delivery: To achieve desired outcomes, institutions implement

Syllabus, teaching methods, activities, and assessment tools.

Continuous Improvement (Middle Arrow):

Results from assessments are **fed back** into the system for continuous improvement in curriculum, teaching strategies, and evaluation methods.

This framework **aligns education with industry needs** and fosters **continuous enhancement** of learning outcomes.

DLT : Analytical framework

Analytical framework : Combination of , theoretical and a conceptual framework

This framework is designed to structure an analyst's thinking, and to help logical thinking in a systematic manner.

Analytical frameworks are models that help people understand and make sense of information.

DLT analytical framework helps central banks and authorities understand DLT-based payment, clearing, and settlement systems by assessing their benefits and risks.

DLT : Analytical framework

Framework based on four core components:

Scope: understanding the arrangement which includes

- Its functionality and nature of service
- Factors for its effective implementation

Efficiency: Analyzing the arrangement's implications for efficiency

Safety: Analyzing the arrangement's implications for safety

B**roader implications**: Analyzing the arrangement's broader financial market implications

Analytical framework - First component of framework : Understanding the arrangement

What is the functionality and nature of the arrangement?

DLT arrangement typically does one or more of the following:

Record Keeping:

It keeps track of information, like who owns an asset or how much money is in an account.

Example: A DLT ledger could record that Alice owns 10 shares of a company.

Transfer of Assets or Updating Balances:

It helps move assets (like money or property) from one person to another and updates the balances automatically.

Example: If Alice has 10 shares and sends 5 shares to Bob, ledger updates to show Alice now has 5 shares and Bob has 5 shares.

Automation Tools:

It uses smart contracts to automate tasks, like releasing payment when certain conditions are met.

Example: A smart contract could automatically pay a supplier when goods are delivered.

DLT : Analytical framework

First component of framework : Understanding the arrangement

What is the functionality and nature of the arrangement?

In understanding the functionality and nature of the service, it is useful to understand the

1. Improvement the arrangement is trying to effect
2. what part (or parts) of the value chain is affected
3. how it is designed and
4. which participants and users will be impacted by it.

First component of framework : Understanding the arrangement

What is the functionality and nature of the arrangement?

1. Identifying problems, inefficiencies or improvements that it is addressing

DLT system is typically introduced to solve a problem or make an existing process better.

Simplifying Processes:

Instead of relying on manual paperwork for financial transactions, DLT automates and speeds up process.

Improving Information Flow: Ensuring all relevant parties have access to the same data in real time.

Reducing Costs:

Removing need for intermediaries (e.g., banks/brokers), thus cutting operational expenses.

Enhancing Financial Inclusion:

Allowing people(especially in remote or underdeveloped areas) without traditional bank accounts to access financial services using blockchain technology.

Increasing Automation: Reducing human errors and manual intervention with smart contracts.

First component of framework : Understanding the arrangement

What is the functionality and nature of the arrangement?

1. Identifying problems, inefficiencies or improvements that it is addressing

Short Examples for Each Case

Simplifying Processes – **Blockchain-based property registry** automates real estate transactions, removing manual paperwork and speeding up ownership transfers.

Improving Information Flow – **Blockchain in supply chains** allows real-time tracking of goods, enabling quick recalls and improved transparency.

Reducing Costs – **Crypto-based remittance services (e.g., Ripple/XRP)** enable instant, low-cost international money transfers by eliminating banks and middlemen.

Enhancing Financial Inclusion – **Decentralized finance (DeFi) platforms** allow users to store, send , and receive digital assets using a mobile phone and a blockchain wallet, providing **financial access without needing a traditional bank account**.

Increasing Automation – **Smart contracts in insurance** automatically process claims (e.g., flight delay compensation) based on real-time data, reducing manual verification

First component of framework : Understanding the arrangement

What is the functionality and nature of the arrangement?

2. Identifying the affected part or parts of the value chain

Impact of an arrangement on value chain("journey" a product or service takes from start to finish, with each step adding value to meet customer needs.) depends on which part is affected and whether it introduces disruptive or incremental innovation

The value chain can be categorized in a number of ways, including:

- Customer identification
- Pre-transaction
- Clearing
- Settlement
- Post-settlement

Now explore each one by one --

First component of framework : Understanding the arrangement

What is the functionality and nature of the arrangement?

2. **Identifying the affected part or parts of the value chain**

Customer Identification – Verifying user identity, ensuring compliance with laws like Know Your Customer (KYC) and Anti-Money Laundering (AML) regulations.

Pre-Transaction: This includes creating, validating, and sending payment instructions or other obligations. It also involves checking asset holdings and preparing data for clearing and settlement.

Clearing: This involves confirming, reconciling, and sometimes netting transactions to determine final positions for settlement.

Settlement – Transferring money or assets to complete a transaction based on agreed terms.

Post-Settlement: This includes activities like reconciliation, reporting, asset servicing (e.g., handling interest payments), and enforcing contract terms (e.g., through smart contracts).

Some arrangements may only affect one or two steps. For example, certain DLT systems focus on improving clearing by sharing information more efficiently but don't handle settlement. Others may transform entire process, enabling not just information exchange but also transfer of assets on a digital ledger.

First component of framework : Understanding the arrangement

What is the functionality and nature of the arrangement?

3. **Understanding the design, technology and associated processes behind an arrangement**

Every **DLT-based system** is designed differently based on **what problem it solves, the technology used, and how it operates**. These differences depend on:

- **The type of service it provides**
- **The technology behind it**
- **How it fits into the existing financial or business system**

1. Banking & Payments (Technology for Faster Transactions)- Example: UPI vs. Blockchain Payments

2. Stock Market (Automation for Faster Settlements)-- Example: Traditional vs. Blockchain-Based Settlement

3. Supply Chain (Tracking Goods with Blockchain)-- Example: Tracking Mango Exports from India

4. Land Records (Preventing Fraud & Duplication)-- Example: Land Registry in Telangana

DLT systems vary by purpose—some enhance existing processes (e.g., stock settlements), while others introduce new methods (e.g., blockchain land records). Understanding design, technology, and processes helps in selecting the right solution for each industry.

First component of framework : Understanding the arrangement

What is the functionality and nature of the arrangement?

4. Identifying the affected market participants

Understanding which market participants are impacted by a **DLT-based system** is crucial for assessing its effects on **efficiency, security, and risk** in the financial system.

Potential categories of market participants to consider include:

1 Financial Market Infrastructures (FMIs) – Organizations that handle transactions (e.g., stock exchanges)

2. Banks & Financial Institutions – Entities providing financial services that may benefit or face challenges from DLT.

3 Customers & Businesses – End-users who gain faster, more secure transactions or experience changes in service.

4 Regulatory Authorities – Government bodies ensuring compliance, security, and risk management.

Example: Blockchain in Banking

Before DLT: Banks take 2–3 days to process cross-border transactions due to manual verification.

After DLT: Using blockchain (e.g., **ICICI Bank's blockchain pilot**), transactions are settled **instantly** with reduced costs and improved transparency.

Impact: Banks benefit from **faster settlements**, but regulators must **ensure security and compliance**.

First component of framework : Understanding the arrangement

What are the key factors for effective implementation?

Adoption Requirements:

- Some arrangements involve only a single entity or a small group.
- Others need widespread industry adoption to achieve efficiency gains, as benefits often depend on n/w effects
- A critical mass of participants may be necessary to realize potential benefits.

Market Changes:

Implementation may require significant structural changes, such as shifts in market conventions or practices.

Factors Affecting Implementation:

Environmental: Regulatory, legal, or geopolitical conditions.

Technological: Availability of technology or infrastructure.

Financial: Costs, funding, or economic viability.

***Example*:** A new digital payment system may only succeed if enough banks and customers adopt it, and if existing market practices (e.g., cash reliance) are adjusted to accommodate it.

Analytical framework- First component of framework : Understanding the arrangement

What are the key factors for effective implementation?

Following factors plays important part in the implementation of an arrangement:

1. Environmental Factors

These are external conditions that influence the adoption of a new arrangement. They include:

- **Institutional acceptance**: Whether organizations are willing to adopt new technology.
- **Market factors**: The size of the market, its structure, and existing practices.
- **Regulatory and legal conditions**: Rules and laws that may support or hinder adoption.
- **Industry coordination**: How well different players in the industry work together.

Example:

A new blockchain-based payment system might face challenges if banks are hesitant to adopt it (institutional acceptance), if the market is dominated by cash transactions (market practices), or if regulations are unclear about digital currencies (regulatory conditions).

Analytical framework- First component of framework : Understanding the arrangement

What are the key factors for effective implementation?

Following factors plays important part in the implementation of an arrangement:

2. Technological Factors

These relate to the technology itself and how well it fits into existing systems. Key considerations include:

- **Maturity of the technology**: Is the technology proven and reliable, or still experimental?
- **Interoperability**: Can it work seamlessly with current systems and processes?

Example:

A new AI-powered trading platform is more likely to be adopted if it is compatible with existing trading software (interoperability) and has been tested successfully in other markets (maturity)

Analytical framework- First component of framework : Understanding the arrangement

What are the key factors for effective implementation?

Following factors plays important part in the implementation of an arrangement:

3. Financial Factors

These involve economic benefits of adopting arrangement. Projects are more likely to be adopted if they:

- Offer **cost savings**: Reduce operational expenses.
- Have **revenue potential**: Generate new income streams.

Example:

A bank may adopt a new cloud-based system if it reduces IT maintenance costs (cost savings) and allows the bank to offer new digital services to customers (revenue potential).

By considering all these three (Environmental , Technological and Financial) factors, organizations can better evaluate whether a new arrangement is feasible and beneficial.

Environmental Factors (Regulatory & Market Acceptance)

- Land Registry on Blockchain – Governments must approve blockchain for official records.
- Decentralized Identity Verification – Industries need common standards for wide adoption.
- Cross-Border Payments – Regulatory compliance is necessary for legal transactions.

Technological Factors (Integration & Compatibility)

- Blockchain Supply Chain Tracking – Must connect with logistics software for real-time updates.
- Smart Contracts Across Blockchains – Needs interoperability to function on different networks.
- DLT-Based Healthcare Records – Must work with hospital databases while ensuring data privacy.

Financial Factors (Cost & Benefits)

- DLT-Based Remittance Services – Should reduce transaction fees to compete with banks.
- Tokenized Real Estate Investment – Needs to offer better returns to attract investors
- DLT in Insurance Claims – Should lower fraud risks and improve claim processing efficiency.

Second component of framework : Potential implications for efficiency

Many believe that DLT can improve market efficiency. Efficiency here refers to how well the system works, including its design, functionality, and resource requirements.

Under this point we will explore 5 points

1. Speed of end-to-end processing
2. Cost of processing
3. Speed and transparency in reconciliation
4. Cost of credit and liquidity management
5. Efficiency gains from automated contract tools

Efficiency is about speed, cost, and effectiveness of a system and it depends on reconciliation (means matching records to ensure accuracy) **, liquidity management, and automation**.

Example:

A student who scores well, meets deadlines, and balances studies with extracurricular is efficient

A DLT-based stock trading speed up transactions (from days to seconds), reduce costs (by cutting out middlemen), and improve accuracy (through automated reconciliation)

Analytical framework: Potential implications for efficiency

Second component of framework : Potential implications for efficiency

Under this point we will explore 5 points –Let-us explore first point

1. **Speed of end-to-end processing-** How to speed-up end-to-end processing so efficiency will improve

- **DLT can speed up transactions** by reducing delays in **information sharing** among participants.
- It helps **simplify processes**, making settlements **more efficient** at an **ecosystem level** (across entire value chain).
- However, **within the system itself**, DLT transactions might be **slower** than traditional methods due to the **complex validation process**.

Example:

RTGS (Real-Time Gross Settlement): Used by banks for high-value transactions, where payments are settled **instantly** by a central authority (RBI in India).

DLT-Based Settlement: Transactions go through **decentralized validation** (multiple nodes approving), which may take longer than RTGS.

DLT improves efficiency overall, but its internal processing may sometimes be slower than centralized systems like RTGS due to its consensus mechanism

Second component of framework : Potential implications for efficiency

Under this point we will explore 5 points –Let-us explore second point

2. **Cost of processing** :

Industry may explore variety of DLT arrangements to reduce costs in certain parts of the value chain

For example

- DLT could **reduce costs** by sharing maintenance expenses among participants instead of a central institution handling all the work and charging fees
- DLT eliminates intermediaries like SWIFT, reducing transaction fees.
- Smart contracts automate **letter of credit** issuance, reducing paperwork and manual verification costs.
- Hospitals can **share verified patient records**, reducing administrative expenses.
- DLT removes the need for **brokers and legal intermediaries**, lowering transaction costs.

DLT lowers costs by eliminating middlemen and automating processes, but some users bear higher direct expenses for infrastructure and maintenance.

Second component of framework : Potential implications for efficiency

Now explore 3rd point-- Speed and Transparency in Reconciliation

What is Reconciliation?

- Ensuring that internal transaction records match across relevant parties.
- Traditionally a **time-consuming and labor-intensive** process.
- Involves reconciling information from different ledgers, recording, and storing it in various formats.

How DLT Improves Reconciliation?

Faster Processing: All parties access a **shared, consistent record**, avoiding manual comparison.

Error Reduction: Minimizes discrepancies and back-office work.

Transparency: Data can be shared across participants for better **market visibility**.

Privacy Concerns: While improving reconciliation speed, **data protection and privacy must be ensured**.

Example:

Banking Sector: **State Bank of India (SBI)** is exploring blockchain to reconcile transactions across branches, reducing errors and settlement time.

Stock Markets: **National Stock Exchange (NSE) of India** is piloting DLT for clearing and settlement, ensuring faster trade reconciliation.

Insurance Claims: **ICICI Lombard** uses blockchain to speed up claim verification, reducing fraud and delays.

Second component of framework : Potential implications for efficiency

Under this point we will explore 5 points –Let-us explore fourth point

4. **Cost of credit and liquidity management :**

Credit --Borrowed money, and **liquidity** -- Cash available to make payments.

Faster Transactions Reduce the Need for Credit

- In traditional banking, transactions take time to complete, so businesses often borrow money (credit) to cover delays
- DLT enables **instant or faster payments**, reducing the need to borrow money.

Higher Liquidity Requirements

- Since transactions happen **instantly/near-instantly**, participants must have **enough liquidity** at all times.
- Unlike traditional systems where delays allow firms to manage cash flows, **DLT-based instant settlement requires immediate funds**.

Reduced Collateral Requirements

- DLT reduces need for collateral, so businesses have more money available instead of keeping it locked for security as in traditional banking.

Second component of framework : Potential implications for efficiency

Under this point we will explore 5 points –Let-us explore fifth point

5.Efficiency gains from automated contract tools :

Smart contracts are automatic agreements that run on DLT .They follow pre-set rules and execute actions on their own without needing a person to approve them. This makes transactions faster and more efficient.

Examples:

- **Automatic Loan Payments** – A DLT-based loan contract can automatically deduct payments from a borrower's account on the due date, without needing a bank employee.
- **Instant Insurance Claims** – Blockchain-based insurance system can auto-approve claims if all required documents are verified, reducing delays.

Why It's Useful?

- Saves time by removing manual approvals.
- Reduces errors since there's no human involvement.
- Makes transactions faster and more efficient.

Conclusion:

Smart contracts automate processes, making payments, approvals, and transactions faster and hassle-free.

Third component of framework : Potential implications for safety

Under this point we will explore 5 points

1. Operational and security risk

- Reliability & Resilience
- Security
- Operational Capacity and Scalability

2. Settlement issues

- Settlement asset
- Operational settlement
- Legal settlement finality

3. Legal risk

4. Governance

5. Data management and protection

DLT can enhance security and efficiency in financial transactions but also introduces risks. If not properly managed, these risks could disrupt the financial system and spread instability.

Third component of framework : Potential implications for safety

Under this point we will explore 5 points – **Lets-us explore first point Operational and Security Risk:**

Refers to dangers and challenges a system faces while running and keeping data safe.

Operational Risk: Problems that affect how the system works, like system failures, slow processing, or handling too many transactions at once.

Security Risk: Threats like hacking, data theft, or weak encryption that could allow attackers to manipulate or steal information.

In simple terms, it's about making sure the system runs smoothly and stays protected from cyber threats.

This Operational and Security Risk covers three points

- Reliability & Resilience
- Security
- Operational Capacity and Scalability

Third component of framework : Potential implications for safety

Reliability & Resilience: One of key drivers of DLT implementation is its potential to strengthen an arrangement's resilience and reliability

DLT improves system strength by storing data on multiple computers. If one fails, others keep working, preventing system failure. This helps protect against cyber threats. However, more computers also mean more chances for hackers to attack.

Security: DLT relies on cryptographic tools like public key cryptography to keep data safe. While these tools are strong today, future technology may weaken them. A poorly managed system may struggle to handle new security threats. Transitioning from old systems to DLT may also create security risks, so multiple security layers should be used

Operational Capacity and Scalability : A payment system must handle large and fluctuating transaction volumes, especially during busy periods. If it cannot scale properly, it may become unsafe. Scalability depends on data storage, consensus mechanisms, and system design.

Third component of framework : Potential implications for safety

Now explore the second point , **Settlement issues** :

DLT can speed up financial transactions, making payments and settlements more efficient. However, it also impacts safety. Key factors include the type of asset being settled, how transactions are processed, and when they become legally final.

Example : Imagine you buy a car using cryptocurrency.

Settlement Asset: Digital payment (crypto) and car ownership record.

Operational Settlement: Blockchain updates transaction, showing seller received payment.

Legal Finality: Once confirmed on blockchain, ownership transfer is final, preventing fraud or disputes.

Following three points are important fro settlement issues point of view

- Settlement asset
- How settlement is achieved operationally and
- How settlement finality is achieved for legal purposes.

Settlement Asset

Settlement Asset is what is being recorded or transferred in a transaction.

Balance Update: System records changes in account balances (debits and credits).

Digital Asset Transfer: Ownership of a digital asset (like cryptocurrency) is recorded on the ledger.

Physical Asset Representation: Ledger tracks digital versions of real-world assets (like gold stored in a vault).

Example:

If you transfer ₹1,000 to a friend, the bank updates your balance (balance update).

If you send Bitcoin, ownership of the Bitcoin is recorded (digital asset transfer).

If you trade gold-backed tokens, ledger tracks ownership while gold remains in a vault (physical asset representation).

Operational Settlement

Updating and confirming transactions in a system, ensuring all participants agree before they are final.

In DLT, transactions are not final immediately—they take time to update and sync across multiple computers

Consensus Delay: System must confirm transaction, which may take time.

Proof-of-Work Model: More confirmations make transaction more secure and less likely to be reversed

Complex Transactions: If two assets are exchanged (e.g., stocks for money or one currency for another), settlement becomes more complicated and may involve other financial systems.

Example:

If you send Bitcoin, transaction appears quickly but is only fully secure after multiple confirmations on blockchain.

Legal Settlement Finality

Settlement finality means a transaction is legally complete and cannot be reversed, even if one party goes bankrupt.

Traditional Systems: Settlement finality happens at a clear, legally defined moment.

DLT Systems: Finality depends on the consensus mechanism, which may not have a fixed point of completion. Some legal systems may not fully recognize DLT-based settlements.

Example:

In a bank transfer, once money reaches the recipient's account, it cannot be reversed (finality is clear). In Bitcoin, a transaction becomes more final as more blocks confirm it, but there is no single, legally defined finality point.

Third component of framework : Potential implications for safety

Under this point we will explore 5 points – **Now explore the third point Legal risk** :

Legal risk arises when there is uncertainty about laws governing financial transactions using DLT.

Since DLT is new, existing laws may not fully apply, creating confusion about ownership, contracts, and settlement rules.

However, DLT can also reduce legal risks by automating contract terms to ensure they are followed correctly.

Example:

If you buy property using blockchain, but law does not recognize blockchain records as proof of ownership, legal disputes may arise.

On other hand, smart contracts can automate rent payments, ensuring timely enforcement of agreements.

Third component of framework : Potential implications for safety

Under this point we will explore 5 points – Now explore the Forth point **Governance :**

It refers to rules and decision-making processes that manage a DLT system.

A strong governance structure improves security and efficiency, while a weak or complex one can slow responses to problems. Clear rules are needed for managing changes and handling issues.

Example:

In a blockchain-based voting system, clear governance ensures fair rule changes and quick fixes for security issues. Without proper governance, resolving disputes or updating the system could take too long, causing disruptions.

Third component of framework : Potential implications for safety

Under this point we will explore 5 points –

Now explore the Fifth point **Data management and protection**

Data in a DLT system must be secure, accurate, and traceable.

It should not be lost, changed, or misused. While transparency is important, privacy must also be protected by controlling who can see or edit the data.

Example:

In a blockchain-based bank system, all transactions are recorded permanently (immutability).

However, if a wrong transaction occurs due to fraud or error, there must be a way to correct it while maintaining security and trust.

Fourth component of framework : Potential broader financial market implications

It means impact of changes in one financial system on other connected systems, like banks, payments, and global markets.

As DLT grows, it can impact entire financial system since banks, payment systems, and securities markets are all connected. Many financial transactions involve multiple systems, sometimes across different countries, making DLT's role even more significant.

Example: If a bank adopts DLT for faster payments, it can improve stock trading and international transactions. But if the system fails, it could disrupt multiple financial services.

Under this point we will explore 3 points

- Connectivity issues and standards development
- Financial market architecture
- Broader financial market risks

Fourth component of framework : Potential broader financial market implications

Let-us explore first point : Connectivity issues and standards development

1. Connectivity issues and standards development :

As different DLT systems are developed, they may have similar or complementary functions but may not be able to communicate with each other or with traditional financial systems.

This creates a challenge in conducting smooth financial transactions.

To solve this, **technical interoperability standards** are needed.

These standards act as common rules that allow different DLT systems to connect, reducing costs and improving efficiency.

Example:

If two banks use different blockchain platforms for payments, they need a common standard to transfer money smoothly between them, just like different phone networks allowing users to call each other.

Fourth component of framework : Potential broader financial market implications

Now explore 2nd point : **Financial market architecture**

2. **Financial market architecture :**

DLT can impact financial markets in different ways:

Small Changes (Incremental Upgrade): Some DLT systems improve existing financial processes without major disruptions.

Big Changes (Disintermediation): Some DLT systems remove the need for middlemen (like banks), changing how financial transactions happen and affecting competition

New Players: DLT may allow new non-bank companies to enter financial markets, but they may not be covered by existing regulations.

Example:

If banks use DLT to process payments faster, it's just an improvement.

But if people use blockchain-based systems to transfer money without banks, it changes how the financial market works and introduces new, unregulated competitors.

Fourth component of framework : Potential broader financial market implications

Now we will explore 3rd point : **Broader financial market risks**

3. **Broader financial market risks:**

DLT can both reduce and increase financial risks:

Positive Impact: DLT improves transparency by sharing data across financial systems, helping institutions manage risks better.

Negative Impact: Too much automation in financial markets could trigger unexpected problems.

For example, if economic conditions worsen, automated margin calls (requests for more funds in investments) could happen at the same time across multiple systems, leading to a sudden liquidity crisis.

Example:

Imagine a stock market using DLT, where smart contracts automatically sell shares if prices drop. If too many investors experience this at once, it could cause panic selling, making the market crash faster. Therefore, safeguards are needed to prevent such risks.

Thank You